

Experiment/ Case Study No.:

1. **Experiment Title:** Demonstration of var, let, const, Template Literals, Destructuring and Billing Calculator using JavaScript

2. **Software/Tools Required:** VS Code, Web Browser, HTML, CSS, JavaScript

3. **Experiment Program Code:**

product.html

```
<!DOCTYPE html>
<html>

<head>
  <meta charset="UTF-8">
  <title>Product Calculation</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #e8f5e9;
      margin: 50px auto;
      max-width: 400px;
    }

    .container {
      background: #ffffff;
      padding: 20px;
      border: 2px solid #4caf50;
      border-radius: 8px;
    }

    h2 {
      text-align: center;
      color: #2e7d32;
    }

    .form-group {
      margin-bottom: 15px;
    }

    label {
      display: block;
      margin-bottom: 5px;
      font-weight: bold;
      color: #333;
    }

    input {
      width: 100%;
      padding: 8px;
      box-sizing: border-box;
      border: 1px solid #ccc;
      border-radius: 4px;
    }

    input:focus {
      border-color: #4caf50;
      outline: none;
    }

    button {
      width: 100%;
      padding: 10px;
      background-color: #4caf50;
      color: white;
      border: none;
      border-radius: 4px;
      font-size: 16px;
      cursor: pointer;
    }

    button:hover {
      background-color: #388e3c;
    }

    #result {
      margin-top: 20px;
      padding: 15px;
      background-color: #c8e6c9;
      border: 1px solid #4caf50;
      border-radius: 4px;
      display: none;
    }

    .result-item {
      display: flex;
      justify-content: space-between;
      margin-bottom: 5px;
    }
```

```

    }

    hr {
        border: 0;
        border-top: 1px solid #4caf50;
        margin: 10px 0;
    }
</style>
</head>

<body>
    <div class="container">
        <h2>Product Invoice</h2>
        <div class="form-group">
            <label for="customerName">Customer Name</label>
            <input type="text" id="customerName" placeholder="Enter customer name">
        </div>
        <div class="form-group">
            <label for="product">Product Name</label>
            <input type="text" id="product" placeholder="Enter product name">
        </div>
        <div class="form-group">
            <label for="price">Price</label>
            <input type="number" id="price" placeholder="Enter price" min="0">
        </div>
        <div class="form-group">
            <label for="quantity">Quantity</label>
            <input type="number" id="quantity" placeholder="Enter quantity" min="1" value="1">
        </div>
        <button onclick="calculateBill()">Calculate Bill</button>

        <div id="result">
            <div class="result-item"><span>Customer:</span> <strong id="res-customer"></strong></div>
            <div class="result-item"><span>Product:</span> <strong id="res-product"></strong></div>
            <div class="result-item"><span>Subtotal:</span> <strong id="res-subtotal"></strong></div>
            <div class="result-item"><span>GST (18%):</span> <strong id="res-gst"></strong></div>
            <hr>
            <div class="result-item"><span>Total:</span> <strong id="res-total"></strong></div>
        </div>
    </div>

    <p style="margin-top: 20px; color: #2e7d32; font-weight: bold;">Made by Ayush Bisen</p>
    <p style="text-align: center;"></p>

    <script src="script.js"></script>
</body>
</html>

```

script.js

```

function calculateBill() {
  // let example
  let customerName = document.getElementById("customerName").value;
  let product = document.getElementById("product").value;
  let price = Number(document.getElementById("price").value);
  let quantity = Number(document.getElementById("quantity").value);

  if (!customerName || !product || price <= 0 || quantity <= 0) {
    alert("Please enter valid details for all fields.");
    return;
  }

  const GST = 0.18;

  let subtotal = price * quantity;
  let gstAmount = subtotal * GST;
  let total = subtotal + gstAmount;

  const bill = {
    customerName,
    product,
    subtotal,
    gstAmount,
    total
  };

  document.getElementById("res-customer").textContent = bill.customerName;
  document.getElementById("res-product").textContent = bill.product;
  document.getElementById("res-subtotal").textContent = bill.subtotal.toFixed(2);
  document.getElementById("res-gst").textContent = bill.gstAmount.toFixed(2);
  document.getElementById("res-total").textContent = bill.total.toFixed(2);

  document.getElementById("result").style.display = "block";

  console.log("Calculation completed.");
  console.log("Bill generated for:", bill.customerName);
  console.log("Total Amount:", bill.total);
}

```

4. Output:



5. Case Study Title: Billing Calculator Webpage for a Shopping App using JavaScript, HTML, and CSS

6. Case Study Program Code:

shopping.html

```

<!DOCTYPE html>
<html>

<head>
  <meta charset="UTF-8">
  <title>Online Shopping Bill Calculator</title>
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome@6.4.0/css/all.min.css">
  <style>
    :root {
      --primary-green: #2e7d32;
      --light-green: #e8f5e9;
      --hover-green: #388e3c;
      --accent-green: #4caf50;
      --dark-green: #1b5e20;
      --bg-color: #f1f8e9;
      --text-color: #333;
    }

    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background-color: var(--bg-color);
      margin: 0;
      padding: 20px;
      color: var(--text-color);
    }

    .container {
      max-width: 1200px;
      margin: 0 auto;
      background: #ffffff;
      padding: 30px;
      border-radius: 12px;
      box-shadow: 0 4px 15px rgba(46, 125, 50, 0.1);
      border-top: 5px solid var(--primary-green);
    }

    h1,
    h2 {
      text-align: center;
    }
  </style>

```

```

        color: var(--dark-green);
        margin-top: 0;
    }

    .input-section {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
        gap: 15px;
        margin-bottom: 20px;
    }

    .form-group {
        display: flex;
        flex-direction: column;
    }

    label {
        margin-bottom: 5px;
        font-weight: 600;
        color: var(--primary-green);
    }

    input {
        padding: 10px;
        border: 1px solid #c8e6c9;
        border-radius: 6px;
        font-size: 14px;
        transition: all 0.3s;
    }

    input:focus {
        border-color: var(--accent-green);
        box-shadow: 0 0 5px rgba(76, 175, 80, 0.3);
        outline: none;
    }

    input:disabled {
        background-color: #f5f5f5;
        cursor: not-allowed;
    }

    button {
        padding: 12px 20px;
        background-color: var(--accent-green);
        color: white;
        border: none;
        border-radius: 6px;
        font-size: 16px;
        font-weight: bold;
        cursor: pointer;
        transition: background-color 0.3s, transform 0.1s;
    }

    button:hover {
        background-color: var(--hover-green);
    }

    button:active {
        transform: scale(0.98);
    }

    button.secondary {
        background-color: var(--primary-green);
    }

    button.secondary:hover {
        background-color: var(--dark-green);
    }

    .actions {
        display: flex;
        gap: 10px;
        justify-content: center;
        margin-bottom: 30px;
    }

    table {
        width: 100%;
        border-collapse: collapse;
        margin-bottom: 20px;
    }

    th,
    td {
        padding: 12px;
        text-align: left;
    }

```

```

        border-bottom: 1px solid #c8e6c9;
    }

    th {
        background-color: var(--light-green);
        color: var(--dark-green);
        font-weight: 600;
    }

    tr:hover {
        background-color: #f1f8e9;
    }

    .main-layout {
        display: flex;
        gap: 30px;
        align-items: flex-start;
    }

    .left-panel {
        flex: 1;
    }

    .right-panel {
        flex: 1;
        background: #f9fbf9;
        padding: 20px;
        border-radius: 8px;
        border: 1px solid #c8e6c9;
    }

    @media (max-width: 768px) {
        .main-layout {
            flex-direction: column;
        }
    }

    .product-grid {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
        gap: 20px;
        margin-bottom: 30px;
    }

    .product-card {
        background-color: white;
        border: 1px solid var(--accent-green);
        border-radius: 8px;
        padding: 15px;
        text-align: center;
        box-shadow: 0 2px 5px rgba(0, 0, 0, 0.05);
        transition: transform 0.2s;
    }

    .product-card:hover {
        transform: translateY(-3px);
        box-shadow: 0 4px 10px rgba(46, 125, 50, 0.15);
    }

    .product-card h3 {
        color: var(--primary-green);
        margin-top: 0;
        margin-bottom: 10px;
    }

    .product-card .price {
        font-size: 18px;
        font-weight: bold;
        color: var(--dark-green);
        margin: 10px 0 15px 0;
    }

    .product-card .add-section {
        display: flex;
        gap: 10px;
        justify-content: center;
    }

    .product-card input {
        width: 60px;
        padding: 8px;
        text-align: center;
    }

    .product-card button {
        padding: 8px 15px;
    }

```

```

    }

    .invoice-section {
      background-color: var(--light-green);
      padding: 20px;
      border-radius: 8px;
      border: 1px dashed var(--accent-green);
      display: none;
    }

    .invoice-header {
      display: flex;
      justify-content: space-between;
      margin-bottom: 20px;
      border-bottom: 2px solid var(--primary-green);
      padding-bottom: 10px;
    }

    .summary-row {
      display: flex;
      justify-content: space-between;
      margin-bottom: 8px;
      font-size: 16px;
    }

    .summary-row.total {
      font-size: 20px;
      font-weight: bold;
      color: var(--dark-green);
      border-top: 2px solid var(--accent-green);
      padding-top: 10px;
      margin-top: 10px;
    }

    .delete-btn {
      background-color: #ef5350;
      padding: 6px 12px;
      font-size: 12px;
    }

    .delete-btn:hover {
      background-color: #d32f2f;
    }

    @media print {
      body {
        background-color: white;
        padding: 0;
      }

      .container {
        box-shadow: none;
        border: none;
        padding: 0;
      }

      .no-print {
        display: none !important;
      }

      .invoice-section {
        display: block !important;
        border: none;
        background-color: transparent;
        padding: 0;
      }
    }

    .footer {
      text-align: center;
      margin-top: 20px;
      color: var(--primary-green);
      font-weight: bold;
    }

    #emptyCartMessage {
      text-align: center;
      color: #81c784;
      font-style: italic;
      padding: 20px;
    }
  }
</style>
</head>

<body>
  <div class="container">

```

```

<h1 class="no-print"><i class="fas fa-leaf"></i> Checkout</h1>

<div class="main-layout">
  <div class="left-panel no-print" id="shopContainer">
    <h2><i class="fas fa-store"></i> Shop Products</h2>
    <div class="product-grid">
      <div class="product-card">
        <h3><i class="fas fa-solar-panel"></i> Solar Panels</h3>
        <p class="price">₹15000.00</p>
        <div class="add-section">
          <input type="number" id="qty-solar" value="1" min="1">
          <button onclick="addToCart('Solar Panels', 15000, 'qty-solar')">Add</button>
        </div>
      </div>
      <div class="product-card">
        <h3><i class="fas fa-bolt"></i> Inverter</h3>
        <p class="price">₹10000.00</p>
        <div class="add-section">
          <input type="number" id="qty-inverter" value="1" min="1">
          <button onclick="addToCart('Inverter', 10000, 'qty-inverter')">Add</button>
        </div>
      </div>
      <div class="product-card">
        <h3><i class="fas fa-plug"></i> Power Cables</h3>
        <p class="price">₹500.00</p>
        <div class="add-section">
          <input type="number" id="qty-cables" value="1" min="1">
          <button onclick="addToCart('Power Cables', 500, 'qty-cables')">Add</button>
        </div>
      </div>
      <div class="product-card">
        <h3><i class="fas fa-battery-full"></i> Battery</h3>
        <p class="price">₹12000.00</p>
        <div class="add-section">
          <input type="number" id="qty-battery" value="1" min="1">
          <button onclick="addToCart('Battery', 12000, 'qty-battery')">Add</button>
        </div>
      </div>
    </div>
  </div>

  <div class="right-panel">
    <div class="no-print">
      <div class="form-group" style="margin-bottom: 20px;">
        <label for="customerName">Customer Name</label>
        <input type="text" id="customerName" placeholder="Enter customer name">
      </div>

      <div class="manual-add-section"
        style="background: #e8f5e9; padding: 15px; border-radius: 8px; margin-bottom: 20px; border: 1px dashed
#4caf50;">
        <h3 style="margin-top:0; margin-bottom:10px; color: var(--primary-green); font-size: 16px;"><i
class="fas fa-plus-circle"></i> Custom Item</h3>
        <div style="display:flex; gap:10px; align-items:flex-end;">
          <div class="form-group" style="flex:2;">
            <label for="product" style="font-size: 13px;">Product</label>
            <input type="text" id="product" placeholder="Item name" style="padding: 8px;">
          </div>
          <div class="form-group" style="flex:1.5;">
            <label for="price" style="font-size: 13px;">Price (₹)</label>
            <input type="number" id="price" placeholder="0" min="0" step="0.01"
style="padding: 8px;">
          </div>
          <div class="form-group" style="flex:1;">
            <label for="quantity" style="font-size: 13px;">Qty</label>
            <input type="number" id="quantity" placeholder="1" min="1" value="1"
style="padding: 8px;">
          </div>
          <button onclick="addItem()"
style="padding: 9px 12px; background-color: var(--primary-green); margin-bottom: 2px;"><i
class="fas fa-cart-plus"></i></button>
        </div>
      </div>
    </div>

    <div id="cartSection" class="no-print">
      <table id="cartTable">
        <thead>
          <tr>
            <th>Product</th>
            <th>Price</th>
            <th>Qty</th>
            <th>Total</th>
            <th>Action</th>
          </tr>
        </thead>
      </table>
    </div>
  </div>

```

```

        <tbody id="cartBody">
        </tbody>
    </table>
    <div id="emptyCartMessage">Your cart is currently empty.</div>

    <div class="actions" id="mainActions" style="margin-top: 20px;">
        <button class="secondary" onclick="generateInvoice()"><i class="fas fa-file-invoice"></i>
            Generate Invoice</button>
    </div>
</div>

<div id="invoiceSection" class="invoice-section">
    <div class="invoice-header">
        <div>
            <h2 style="margin: 0; color: var(--dark-green);"><i class="fas fa-leaf"></i> GreenMart
                Invoice</h2>
            <p style="margin: 5px 0 0 0;" id="invDate"></p>
        </div>
        <div style="text-align: right;">
            <strong>Customer:</strong> <span id="invCustomer"></span><br>
            <strong>Invoice #:</strong> <span id="invNumber"></span>
        </div>
    </div>

    <table id="invoiceTable">
        <thead>
            <tr>
                <th>Product</th>
                <th>Price</th>
                <th>Qty</th>
                <th>Total</th>
            </tr>
        </thead>
        <tbody id="invoiceBody">
            <!-- Invoice items -->
        </tbody>
    </table>

    <div style="width: 100%; max-width: 300px; margin-left: auto; margin-top: 20px;">
        <div class="summary-row">
            <span>Total Items:</span>
            <strong id="res-items">0</strong>
        </div>
        <div class="summary-row">
            <span>Subtotal:</span>
            <strong id="res-subtotal">₹0.00</strong>
        </div>
        <div class="summary-row">
            <span>Tax (18% GST):</span>
            <strong id="res-gst">₹0.00</strong>
        </div>
        <div class="summary-row total">
            <span>Grand Total:</span>
            <strong id="res-total">₹0.00</strong>
        </div>
    </div>

    <div class="actions no-print" style="margin-top: 30px;">
        <button onclick="window.print()"><i class="fas fa-print"></i> Print Invoice</button>
        <button class="secondary" onclick="resetForm()"><i class="fas fa-plus-circle"></i> New
            Transaction</button>
    </div>
</div>
</div>
</div>

<p class="footer no-print">Made by Ayush Bisen</p>
</div>

<script src="shopping.js"></script>
</body>
</html>

```

shopping.js

```

let cart = [];
const GST_RATE = 0.18;

document.addEventListener('DOMContentLoaded', () => {
    updateCartTable();
});

function addItem() {
    const customerNameInput = document.getElementById("customerName");
    const productInput = document.getElementById("product");

```

```

const priceInput = document.getElementById("price");
const quantityInput = document.getElementById("quantity");

const customerName = customerNameInput.value.trim();
const product = productInput.value.trim();
const price = parseFloat(priceInput.value);
const quantity = parseInt(quantityInput.value);

if (!customerName) {
    alert("Please enter customer name.");
    customerNameInput.focus();
    return;
}

if (!product) {
    alert("Please enter product name.");
    productInput.focus();
    return;
}

if (isNaN(price) || price <= 0) {
    alert("Please enter a valid price.");
    priceInput.focus();
    return;
}

if (isNaN(quantity) || quantity <= 0) {
    alert("Please enter a valid quantity.");
    quantityInput.focus();
    return;
}

const itemTotal = price * quantity;

const item = {
    id: Date.now(),
    product,
    price,
    quantity,
    total: itemTotal
};

cart.push(item);

customerNameInput.disabled = true;

// Clear product inputs
productInput.value = "";
priceInput.value = "";
quantityInput.value = "1";
productInput.focus();

updateCartTable();
}

function addToCart(productName, price, qtyInputId) {
    const customerNameInput = document.getElementById("customerName");
    const quantityInput = document.getElementById(qtyInputId);

    const customerName = customerNameInput.value.trim();
    const quantity = parseInt(quantityInput.value);

    if (!customerName) {
        alert("Please enter customer name before adding items.");
        customerNameInput.focus();
        return;
    }

    if (isNaN(quantity) || quantity <= 0) {
        alert("Please enter a valid quantity.");
        quantityInput.focus();
        return;
    }

    const existingItem = cart.find(item => item.product === productName);
    if (existingItem) {
        existingItem.quantity += quantity;
        existingItem.total = existingItem.price * existingItem.quantity;
    } else {
        const itemTotal = price * quantity;
        const item = {
            id: Date.now() + Math.random(),
            product: productName,
            price,
            quantity,
            total: itemTotal
        };
    }
}

```

```

    };
    cart.push(item);
  }

  customerNameInput.disabled = true;
  quantityInput.value = "1";
  updateCartTable();
}

function removeItem(id) {
  cart = cart.filter(item => item.id !== id);
  updateCartTable();

  if (cart.length === 0) {
    document.getElementById("customerName").disabled = false;
    document.getElementById("invoiceSection").style.display = "none";
  }
}

function updateCartTable() {
  const cartBody = document.getElementById("cartBody");
  const cartTable = document.getElementById("cartTable");
  const emptyCartMessage = document.getElementById("emptyCartMessage");

  cartBody.innerHTML = "";

  if (cart.length === 0) {
    cartTable.style.display = "none";
    emptyCartMessage.style.display = "block";
  } else {
    cartTable.style.display = "table";
    emptyCartMessage.style.display = "none";

    cart.forEach(item => {
      const row = document.createElement("tr");
      row.innerHTML = `
        <td>${item.product}</td>
        <td>₹${item.price.toFixed(2)}</td>
        <td>${item.quantity}</td>
        <td>₹${item.total.toFixed(2)}</td>
        <td><button class="delete-btn" onclick="removeItem(${item.id})"><i class="fas fa-trash"></i></button></td>
      `;
      cartBody.appendChild(row);
    });
  }
}

function generateInvoice() {
  if (cart.length === 0) {
    alert("Cart is empty. Please add items to generate invoice.");
    return;
  }

  const customerName = document.getElementById("customerName").value.trim();

  // Populate invoice details
  document.getElementById("invCustomer").textContent = customerName;
  document.getElementById("invNumber").textContent = "INV-" + Math.floor(100000 + Math.random() * 900000);

  const options = { year: 'numeric', month: 'long', day: 'numeric', hour: '2-digit', minute: '2-digit' };
  document.getElementById("invDate").textContent = "Date: " + new Date().toLocaleDateString('en-IN', options);

  const invoiceBody = document.getElementById("invoiceBody");
  invoiceBody.innerHTML = "";

  let subtotal = 0;
  let totalItems = 0;

  cart.forEach(item => {
    subtotal += item.total;
    totalItems += item.quantity;

    const row = document.createElement("tr");
    row.innerHTML = `
      <td>${item.product}</td>
      <td>₹${item.price.toFixed(2)}</td>
      <td>${item.quantity}</td>
      <td>₹${item.total.toFixed(2)}</td>
    `;
    invoiceBody.appendChild(row);
  });

  const gstAmount = subtotal * GST_RATE;
  const grandTotal = subtotal + gstAmount;

  document.getElementById("res-items").textContent = totalItems;
}

```

```

document.getElementById("res-subtotal").textContent = "₹" + subtotal.toFixed(2);
document.getElementById("res-gst").textContent = "₹" + gstAmount.toFixed(2);
document.getElementById("res-total").textContent = "₹" + grandTotal.toFixed(2);

document.getElementById("invoiceSection").style.display = "block";
document.getElementById("cartSection").style.display = "none";
document.getElementById("shopContainer").style.display = "none";
if (document.querySelector(".manual-add-section")) {
    document.querySelector(".manual-add-section").style.display = "none";
}
document.getElementById("mainActions").style.display = "none";
}

function resetForm() {
    cart = [];
    document.getElementById("customerName").value = "";
    document.getElementById("customerName").disabled = false;
    document.getElementById("product").value = "";
    document.getElementById("price").value = "";
    document.getElementById("quantity").value = "1";

    const qtyInputs = document.querySelectorAll('input[id^="qty-"]');
    qtyInputs.forEach(input => input.value = "1");

    document.getElementById("invoiceSection").style.display = "none";
    document.getElementById("cartSection").style.display = "block";
    document.getElementById("shopContainer").style.display = "block";
    if (document.querySelector(".manual-add-section")) {
        document.querySelector(".manual-add-section").style.display = "block";
    }
    document.getElementById("mainActions").style.display = "flex";

    updateCartTable();
}

```

8. Output:

9. Result/Conclusion:

The practical was completed successfully. The concepts of var, let, const, template literals, and destructuring were implemented. A billing calculator was developed that accepts user input, performs GST and Grand Total calculations, and displays the final bill accurately using JavaScript.